

The empirical recurrence for OEIS A295982: recovering a conjecture that is not written down

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Abstract

OEIS A295982 records “Empirical recurrence of order 67”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 123$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 67, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 67 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A295982 is “Number of $n \times 5$ 0..1 arrays with each 1 adjacent to 2 or 3 king-move neighboring 1s.”. It has offset 1 and begins

1, 90, 581, 3851, 62702, 649470, 6388097, 74040837, ...

The entry states:

Empirical recurrence of order 67 (see link above)

As of the “Last modified” line on the live entry (Dec 01 2017 15:32:37, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1056$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 123$ of the 1056, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 123$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 246$ exact terms of the model returns order 67 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{67} c_i a(n-i),$$

c_1	10	c_{18}	−4067375609	c_{35}	−3254373672748	c_{52}	−25427730059
c_2	−11	c_{19}	2835314845	c_{36}	−2315920812606	c_{53}	352204961098
c_3	430	c_{20}	−15479236708	c_{37}	4802437987542	c_{54}	5596802498
c_4	−2148	c_{21}	29229697262	c_{38}	2865210661230	c_{55}	−33834544465
c_5	1473	c_{22}	−13360027603	c_{39}	421477368534	c_{56}	−22237132656
c_6	−30846	c_{23}	83937056987	c_{40}	−2918114000777	c_{57}	1039623296
c_7	75050	c_{24}	−148836489894	c_{41}	−2016964642340	c_{58}	5711022247
c_8	−70442	c_{25}	41878138903	c_{42}	−877293758923	c_{59}	−3335365413
c_9	1403977	c_{26}	−280789168118	c_{43}	1675953286588	c_{60}	−164892453
c_{10}	−1697459	c_{27}	647583737576	c_{44}	3485068313349	c_{61}	316035937
c_{11}	5875113	c_{28}	−160748632699	c_{45}	379214917410	c_{62}	67426005
c_{12}	−35224051	c_{29}	220007120493	c_{46}	−2466110475447	c_{63}	−923823
c_{13}	31154951	c_{30}	−1972098652064	c_{47}	−3209490230232	c_{64}	2476206
c_{14}	−165191895	c_{31}	1123410750325	c_{48}	1263488165600	c_{65}	1632838
c_{15}	476070518	c_{32}	1146105067859	c_{49}	2089256419091	c_{66}	267496
c_{16}	−378811477	c_{33}	3158537092434	c_{50}	−318334493118	c_{67}	14144
c_{17}	1928144715	c_{34}	−3257409738398	c_{51}	−786460336812		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 67, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $67 + 4$ terms removed returns order 67 as well, so $\deg q_{\min} = 67 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $67 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 67 that this sequence satisfies — and that family turns out to have exactly one member.

References

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